

- 2.1 Recall the CFG G_4 that we gave in Example 2.4. For convenience, let's rename its variables with single letters as follows.

$$\begin{aligned} E &\rightarrow E + T \mid T \\ T &\rightarrow T \times F \mid F \\ F &\rightarrow (E) \mid a \end{aligned}$$

Give parse trees and derivations for each string.

- | | |
|----------|------------|
| a. a | c. $a+a+a$ |
| b. $a+a$ | d. $((a))$ |

- 2.4 Give context-free grammars that generate the following languages. In all parts, the alphabet Σ is $\{0,1\}$.

- ^Aa. $\{w \mid w \text{ contains at least three 1s}\}$
- b. $\{w \mid w \text{ starts and ends with the same symbol}\}$
- c. $\{w \mid \text{the length of } w \text{ is odd}\}$
- ^Ad. $\{w \mid \text{the length of } w \text{ is odd and its middle symbol is a 0}\}$
- e. $\{w \mid w = w^R, \text{ that is, } w \text{ is a palindrome}\}$
- f. The empty set

- 2.5 Give informal descriptions and state diagrams of pushdown automata for the languages in Exercise 2.4.

- 2.6 Give context-free grammars generating the following languages.

- ^Aa. The set of strings over the alphabet $\{a,b\}$ with more a's than b's
- b. The complement of the language $\{a^n b^n \mid n \geq 0\}$
- ^Ac. $\{w \# x \mid w^R \text{ is a substring of } x \text{ for } w, x \in \{0,1\}^*\}$
- d. $\{x_1 \# x_2 \# \cdots \# x_k \mid k \geq 1, \text{ each } x_i \in \{a,b\}^*, \text{ and for some } i \text{ and } j, x_i = x_j^R\}$

- ^A2.7 Give informal English descriptions of PDAs for the languages in Exercise 2.6.

- 2.11 Convert the CFG G_4 given in Exercise 2.1 to an equivalent PDA, using the procedure given in Theorem 2.20.

- 2.14 Convert the following CFG into an equivalent CFG in Chomsky normal form, using the procedure given in Theorem 2.9.

$$\begin{aligned} A &\rightarrow BAB \mid B \mid \varepsilon \\ B &\rightarrow 00 \mid \varepsilon \end{aligned}$$

- 3.2 This exercise concerns TM M_1 , whose description and state diagram appear in Example 3.9. In each of the parts, give the sequence of configurations that M_1 enters when started on the indicated input string.

- ^Aa. 11.
- b. 1#1.
- c. 1##1.
- d. 10#11.
- e. 10#10.

3.8 Give implementation-level descriptions of Turing machines that decide the following languages over the alphabet $\{0,1\}$.

- ^A**a.** $\{w \mid w \text{ contains an equal number of 0s and 1s}\}$
- b.** $\{w \mid w \text{ contains twice as many 0s as 1s}\}$
- c.** $\{w \mid w \text{ does not contain twice as many 0s as 1s}\}$